AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

MT101
General Mathematics

Midterm Examination (MTA)
Summer Semester 2025/2026

Time Allowed: 90 minutes

Instructions:

- Total Marks 60
- This exam consists of 4 questions.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed

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ESSAY QUESTIONS

Each question is worth 15 marks. You should answer all the questions, giving 60 possible marks.

Q–1: [4+5+6 Marks]

a) perform the indicated operations.

i) multiply: $3i\ (2i-1)^2$

ii) $\frac{5+3i}{1-2i}$. (express the result as a+ bi)

b) Solve for x: -2(4-3x)+7=-2x+6+8x

Q–2: [7+8Marks] solve the following:

a) $|12-4x| > 16$ (write the solution in interval notation)

b) $-2 < -2x+1 < 7$ (write the solution in interval notation)

Q–3: [7+8] Marks] a) solve the following equations:

a) $\frac{2}{3}\sqrt{10x+6}-1=11$

b) $4x^2$ -24x+ 28=0 (use completing the square)

Q–4: [8+7Marks]

a) Find the distance between the points A (8, 3) and the point B(2, -1) and the midpoint of the segment AB.

b) Find the equation of the line that passes through the point (-1, 9) and perpendicular to the line with equation 5x+y = 1

End of the questions.



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